

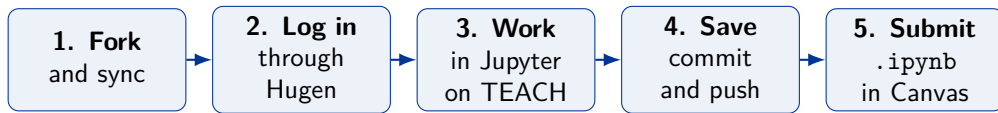
ENGR 1451/2451: Exploratory Data Science

GitHub and Jupyter Workflow through Hugen on the CRCD TEACH Cluster

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Course Workflow



Two places matter

Your work is backed up in your **GitHub fork**; the official assignment is the **notebook submitted in Canvas**.

Hugen and TEACH Are Different

Hugen = website

- Log in at `hugen.crc.pitt.edu`
- Open a TEACH shell
- Launch and manage Jupyter sessions

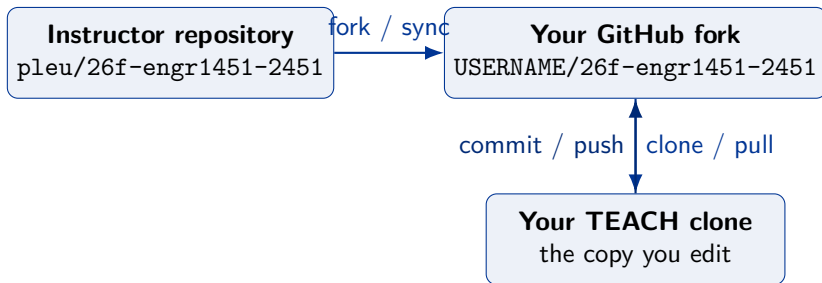
TEACH = cluster

- Jupyter runs on a compute node
- Course software runs here
- Files persist under course storage

Log in through **Hugen**; do the computation on **TEACH**.

Source: CRCDD User Manual, "Hugen / TEACH Open OnDemand Portal."

Three Copies of the Repository



Key idea

Edit the TEACH clone of **your fork**, then push back to **your fork**.

Source: GitHub Docs, "About forks" and "Fork a repository."

Fork Once; Sync Before Each Assignment

Do once

1. Open the instructor repository.
2. Click **Fork**.
3. Keep the repository name unchanged.

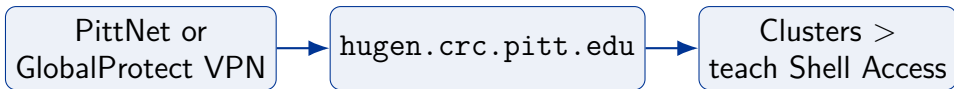
Before new work

1. Open your fork on GitHub.
2. Click **Sync fork**.
3. Click **Update branch** if shown.

Your fork should be at
`github.com/USERNAME/26f-engr1451-2451`

Source: GitHub Docs, "Fork a repository" and "Syncing a fork."

Connect to Hugen



Use the shell for setup

Use the TEACH shell for authentication, file navigation, and Git. Use Jupyter for the assignment analysis.

Source: CRCRD User Manual, "hugen or teach Open OnDemand webportal on the teach cluster."

One-Time GitHub Authentication on TEACH

1. Generate an SSH key:

```
ssh-keygen -t ed25519 -C "PittID@pitt.edu"
```

2. Display and copy the **public** key:

```
cat ~/.ssh/id_ed25519.pub
```

3. GitHub: **Settings** > **SSH and GPG keys** > **New SSH key**.

4. Test the connection:

```
ssh -T git@github.com
```

Keep the private key private

Never copy or share ~/.ssh/id_ed25519; only the file ending in .pub goes to GitHub.

Clone Your Fork Once

Where

Clone inside your personal course folder:

```
~/engr1451
```

What

Clone **your fork**, not the instructor repository.

Check before working

The remote named `origin` should contain your GitHub USERNAME, not `pleu`.

Use the **Git Cheat Sheet** on the next slide for the exact commands.

Source: GitHub Docs, "Cloning a repository."

Launch Jupyter and Work in the Repository

Launch

1. Hugon: **Interactive Apps** > **Jupyter**
2. Use the course account and Python environment.
3. Click **Launch**, then **Connect to Jupyter**.

Work

1. Open engr1451.
2. Enter the cloned repository.
3. Open, run, and save the assigned notebook.

Use relative paths

```
import pandas as pd
df = pd.read_csv("data/Auto.csv")
```

Git Cheat Sheet

One-time setup and clone

```
git config --global user.name "Your Name"
git config --global user.email "PittID@pitt.edu"

cd ~/engr1451
git clone \
  git@github.com:USERNAME/26f-engr1451-2451.git
cd 26f-engr1451-2451
git remote -v
```

Every assignment

```
cd ~/engr1451/26f-engr1451-2451

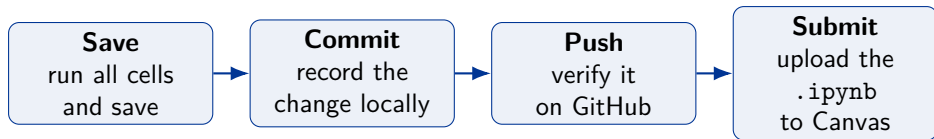
# After syncing your fork on GitHub
git pull origin main

# After finishing the notebook
git status
git diff --stat
git add assignments/assignment01.ipynb
git commit -m "Complete assignment 1"
git push origin main
git log -1 --oneline
```

Remember

Run these commands in the repository. Replace the assignment path and commit message. Add only the files you intend to submit.

From Work to Submission



Finish the cluster session

After saving and pushing, return to **My Interactive Sessions** in Hugen and delete the Jupyter session.

Common Problems

Hugen will not open

Connect to PittNet or GlobalProtect VPN, then reload the portal.

Cannot push

Test SSH and confirm that `origin` points to your fork.

Not a repository

Enter the cloned repository before running Git commands.

Package is missing

Relaunch Jupyter with the course Python environment.

Variables disappeared

Restart from the top and rerun the notebook cells.

References

- CRCSD User Manual. *Hugen / TEACH Open OnDemand portal*.
<https://crc-pages.pitt.edu/user-manual/web-portals/hugen/>
- Pitt Digital. *GlobalProtect VPN Quick Reference*.
<https://services.pitt.edu/TDCClient/33/Portal/KB/Article/3427/>
- GitHub Docs. *Fork a repository and Syncing a fork*.
<https://docs.github.com/en/pull-requests/how-tos/work-with-forks/>
- GitHub Docs. *Connecting to GitHub with SSH*.
<https://docs.github.com/en/authentication/connecting-to-github-with-ssh>